BS2T: Bottleneck Spatial–Spectral Transformer for Hyperspectral Image Classification

Ruoxi Song, Yining Feng, Wei Cheng, Zhenhua Mu, and Xianghai Wang

***Abstract*— Convolutional neural networks (CNNs) have been extensively applied to hyperspectral (HS) image classification tasks and achieved promising performance. However, for CNN-based HS image classification methods, it is hard to depict the dependencies among HS image pixels in long-range distanced positions and bands. Moreover, the limited receptive field of the convolutional layers extremely hinders the development of the CNN structure. To tackle these problems, in this article, the novel bottleneck spatial–spectral transformer (BS2T) is proposed to depict the long-range global dependencies of HS image pixels, which can be regarded as a feature extraction module for HS image classification networks. More specifically, inspired by bottleneck transformer in computer vision, for HS image feature extraction, the proposed BS2T is incorporated with a feature contraction module, a multihead spatial–spectral self-attention (MHS2A) module, and a feature expansion module. In this way, convolutional operations are replaced by the MHS2A to capture the long-range dependency of HS pixels regardless of their spatial position and distance. Meanwhile, in the MHS2A module, to highlight the spectral features of HS images, we introduce the spectral information and content spatial positional information to classical multihead self-attention to make the attention more positional aware and spectral aware. On this basis, a dual-branch HS image classification framework based on 3-D CNN and BS2T is defined for jointly extracting the local–global features of HS images. Experimental results on three public HS image classification datasets show that the proposed classification framework achieves a significant improvement when compared with the state-of-the-art methods. The source code of the proposed framework can be downloaded from https://github.com/srxlnnu/BS2T.**

***Index Terms*— 3-D convolutional neural network (CNN), bottleneck spatial–spectral transformer (BS2T), dual-branch neural network, hyperspectral (HS) image classification, multihead self-attention (MHSA), positional encoding.**

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Ruoxi Song is with the School of Geography, Liaoning Normal University, Dalian 116029, China, and also with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100101, China (e-mail: ruoxisong@qq.com).

Yining Feng and Zhenhua Mu are with the School of Geography, Liaoning Normal University, Dalian 116029, China (e-mail: syfyn@outlook.com; 1161547201@qq.com).

Wei Cheng is with the School of Computer and Information Technology, Liaoning Normal University, Dalian 116029, China (e-mail: chengsolstice@163.com).

Xianghai Wang is with the School of Geography and the School of Computer and Information Technology, Liaoning Normal University, Dalian 116029, China (e-mail: xhwang@lnnu.edu.cn).

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I. Introduction

BY MEASURING of the reflected solar radiation through airborne or spaceborne spectrometers, hyperspectral (HS) images provide the abundant spectral information of land-cover objects, which enables the accurate identification and classification of the underlying land-cover phenomena [1]–[3]. For each HS image pixel, a nearly continuous spectral signature can be obtained over the whole range of wavelengths. The classification of HS images is one of the key procedures in the processing and analysis of HS images, and its purpose is to categorize the content of each HS image pixel within the analyzing scene [4]. The classification of HS images has been applied to many application fields, such as agricultural monitoring [5], [6], natural disaster warning [7], and plant disease detection [8].

In recent years, numerous supervised methods have been proposed for HS image classification, which can be categorized into early stage classification approaches and deep learning-based approaches. Training samples with pixel-level labels are used to train the classifiers [9]. In the early stage HS classification approaches, machine learning methods, such as support vector machine (SVM) algorithm [10], [11], extreme learning machine (ELM) algorithm [12], random forest algorithm [13], and kernel-based strategies [14], have been introduced to the HS image classification. However, these approaches are always designed as "shallow" learning models, which indicates the difficulty in processing nonlinear information when compared with "deep" ones. Meanwhile, the performance of the "shallow" learning methods is greatly dependent on the prior information obtained from the training data, and thus, it is difficult to achieve a balance between classification accuracy with model robustness.

Deep neural networks (DNNs) can extract higher level image features in a data-driven way. In recent years, DNN-based methods have achieved the state-of-the-art (SOTA) performance in a wide range of application fields, such as natural language processing (NLP), image processing, and other real-world applications [15], [16]. Deep learning methods have revolutionized the research field of image analysis and proved to be a powerful tool for processing high-dimensional data [17], which inspired the development of remote sensing image processing such as semantic segmentation, classification [18], and change detection [19]. DNN-based methods, such as stacked autoencoders (SAEs) [20], depth belief networks (DBFs) [21], and recurrent neural networks (RNNs) [22], have

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been successfully applied to HS image classification task. However, only 1-D input is allowed for the abovementioned networks, only spectral information of the HS image pixels is considered. Despite the high spectral resolution, HS images have strong spatial dependencies, which achieves the information complementary to spectral properties [23]. This "joint spatial–spectral" property of HS images requires the HS image classification methods to achieve joint spatial and spectral analyses.

With the unique convolution operations, the convolutional neural networks (CNNs) have shown their potential in feature extraction of image data, and CNN-based HS image classifiers have drawn extensive attention from the research community. Li *et al.* [24] proposed an HS image classification method based on the pixel-level CNN framework, where the hierarchical HS features are automatically extracted in the pixel level. To depict the spatial–spectral HS features in a unified framework, advanced CNN architecture, such as 3-D CNN, and two branch spatial–spectral CNN-based frameworks have been proposed. Li *et al.* [25] proposed a novel 3-D CNN HS image classification framework, where the 3-D convolutional operation is applied to achieve the joint spatial–spectral feature extraction of HS images without any preprocessing or postprocessing processes. Xu *et al.* [26] introduced a two-branch CNN HS image classification framework, where the 1-D CNN branch and 2-D CNN branch simultaneously explores the spectral and spatial information of HS images. On this basis, by introducing the attention mechanism into the classification framework, Li *et al.* [27] presented a dual-branch dual-attention HS image classification framework named double-branch dual-attention mechanism network (DBDA), where 3-D CNN with different kernel sizes is applied to depict the spatial and spectral features and, then, self-attention modules are applied in both of the two branches to highlight the features. Hang *et al.* [28] proposed a novel attention-aided CNN framework for joint spectral–spatial feature extraction and classification of HS images, where the spectral attention and spatial attention are applied to highlight the spatial–spectral features. Similar studies that involve HS image spatial–spectral feature extractors can be found in [29]–[32]. To introduce the content structural information of HS images, Wan *et al.* [33] proposed a context-aware dynamic graph convolutional network, which can flexibly encode the relations among arbitrarily structured non-Euclidean data.

Although the CNN-based HS image classification methods have already achieved promising results, they are still facing the following limitations.

1) *Patch Size Limitation:* The effectiveness of CNN-based HS image feature extractors is completely dependent on the design of the convolution kernel of each neural network layer. Modeling in computer vision has long been dominated by CNNs. However, the CNN architecture has evolved to become increasingly powerful through greater scales, more extensive connections, and more sophisticated forms of convolutions [34]. When the CNN goes deeper, the convolution kernel can extract higher level HS image features. However, since most of the HS image classification methods use spatial–spectral patches as the network input, the size of convolution kernel hinders the development of CNN. Small convolution kernel size is generally selected, which limits the size of the receptive fields, thus influencing the classification result.
2) *Long-Range Dependency:* For existing CNN-based HS image classification methods, it is hard to depict the long-range dependency between HS image pixels with distance positions and bands. Even for networks that involve with self-attention modules, the existing attention mechanism still fails to simultaneously constrain the positional information and spectral information of input HS image patches, which makes the network extremely insensitive to the positional information. Meanwhile, a 2-D CNN method fails to jointly depict the spatial–spectral information of HSI images, while 3-D CNN will involve too many parameters with a high computational cost.

Transformer was first proposed in 2017 and has proven its unique potential in NLP. In transformers, the multihead self-attention (MHSA) modules are used to depict the global long-range dependencies of the input sequences [35]. Recently, transformer has been applied in the field of computer vision [36], [37]. In Vision Transformers (ViTs), positional encoding is equipped with the MHSA to depict the spatial correlation between image pixels [38]. To tackle the abovementioned problems, motivated by ViTs, in this article, a conceptually simple yet powerful bottleneck spatial–spectral transformer (BS2T) is proposed for HS image classification, the proposed transformer can be regarded as a feature extraction module of the classification networks. In BS2T, by introducing the content positional encoding and spectral information to MHSA, the novel multihead spatial–spectral self-attention (MHS2A) is proposed to depict the long-range dependency of HS pixels. On this basis, a dual-branch HS image classification framework based on 3-D CNN and BS2T is proposed for jointly local–global feature extraction of HS images, thus enhancing the classification accuracy.

The main contributions of this article can be concluded as follows.

1) The conceptually simple yet powerful BS2T is proposed for HS image classification.
2) In BS2T, by introducing the content positional coding and spectral information to MHSA, the MHS2A is utilized to depict the long-range dependency of HS image pixels.
3) A dual-branch HS image classification framework based on 3-D CNN and BS2T is proposed for jointly local–global feature extraction of HS images.
4) Experimental results both on randomly selected training datasets and disjoint training datasets demonstrate the effectiveness of the framework. Significant improvement can be observed compared with the SOTA HS classification methods.

The remainder of this article is organized as follows. Section II provides an overview of related works on

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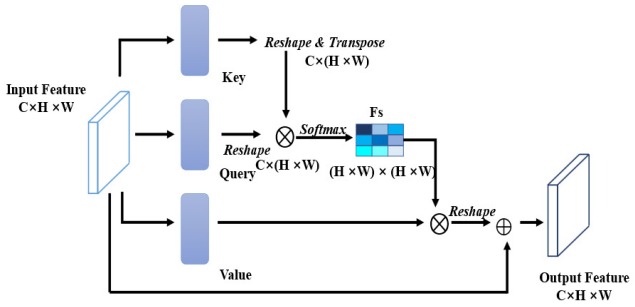


Fig. 1. Self-attention mechanism.

self-attention mechanism and ViTs. The proposed framework is introduced in Section III. In Section IV, experimental setups and results are presented. Conclusions are provided in Section V.

II. Related Work

A. Self-Attention Mechanism

Attention mechanism is oriented from human visual systems. The human visual system tends to use part of the information of the object to make decisions. Similarly, the attention learning mechanism can help the model to dynamically assign different weights to different parts of the input and thus to extract more critical and significant information and guide the model making more appropriate decisions [35], [39], [40]. In the field of artificial intelligence, the attention mechanism has become an integral part of deep learning and has been widely used in NLP, speech recognition, and computer vision tasks.

In general, in computer vision tasks, for an input feature F, the self-attention mechanism can be expressed as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(QK^T\right)V \tag{1}$$

where Q is the query, K is the key, and V is the value. Query is the content to be retrieved, key is the index, and value is the feature to be retrieved. The calculation of attention is to obtain the correlation between query and key, get the attention map, and then get the eigenvalues in value based on the attention map. Fig. 1 shows an illustration of the self-attention mechanism. By highlighting the interested spatial–spectral features of HS image, self-attention mechanism also demonstrated its strong feature discrimination ability in the field of HS image classification [41]–[46]. Mei *et al.* [42]. proposed a novel spectral–spatial attention HS image classification framework, where the spatial and spectral features of HS images are, respectively, extracted by a CNN and an RNN branch, and the spatial and spectral attention mechanism is applied to enhance the features. Sun *et al.* [43]. introduced the self-attention following the spectral modules and spatial modules. In this article, to make the attention operation more position aware and spectral aware, we propose a novel MHS2A equipped with positional encoding to jointly depict the spatial–positional, spatial–spatial, and spatial–spectral correlations among the HS image patch.

B. Transformer in Computer Vision

Transformer does not use RNN sequential structure and convolution operations to depict the input feature correlations, which allows parallelize training for the models. Transformers have proven its irreplaceable potential in NLP [47]. The breakthrough of transformers in NLP tasks sparked great inspiration in the CV community, and research has recently adapted transformers in CV tasks and achieved the SOTA performance in numerous research fields, such as image classification [48], object detection [49], and image super-resolution [50].

The ViT [38] is the first attempt to apply transformer architecture in computer vision tasks. Inspired by transformers in NLP, ViT first divides the original images into blocks; then, the blocks are flattened into sequences and the image pixels are fed into the encoder part of the original transformer model; and finally, ViT inserts a full connection layer to classify the original images. Experiment demonstrates the effectiveness of ViT compared with current SOTA models on large datasets. Very recently, inspired by ViT, He *et al.* [51] introduced the ViT framework to HS image classification. However, in [51], only MHSAs are used to extract the HS spatial–spectral features, HS image local contextual properties are ignored, and thus, the classification accuracy is limited.

III. Methodology

The proposed BS2T net is designed for joint local–global feature extraction of HS images. The framework of the proposed BS2T net is shown in Fig. 2, which consists of three major stages. The first stage extracts the local feature of HS image pixels by 3-D convolutional operation. By defining the spatial–spectral transformer block, the second stage transfers the local information to obtain the global information of HS image pixels. The features obtained from the spectral branch and spatial branch are fused in the last stage. In this section, the BS2T is first proposed to depict the long-range global dependencies of HS image pixels, On this basis, the dual-branch HS image classification framework based on 3-D CNN and BS2T is defined.

A. HS Image Bottleneck Spatial–Spectral Transformer

The overall pipeline of the BS2T is shown in Fig. 2. BS2T can be regarded as a feature extraction module for HS image classification networks, where HS spatial–spectral self-attention with positional encoding is used to extract the global long-range dependencies of HS spatial–spectral features. As shown in Fig. 3, the proposed BS2T is incorporated with a feature contraction module, an MHS2A module, and a feature expansion module.

In a single BS2T block, for each input, HS feature with the size of $H \times W \times C_{in}$ is first fed into the feature contraction module. In the feature contraction module, only Conv Operation with 1×1 kernel size and BN operations are included. After feature contraction, the contracted feature is then fed into the MHS2A module to capture the long-range dependencies of HS images; after that, feature expansion

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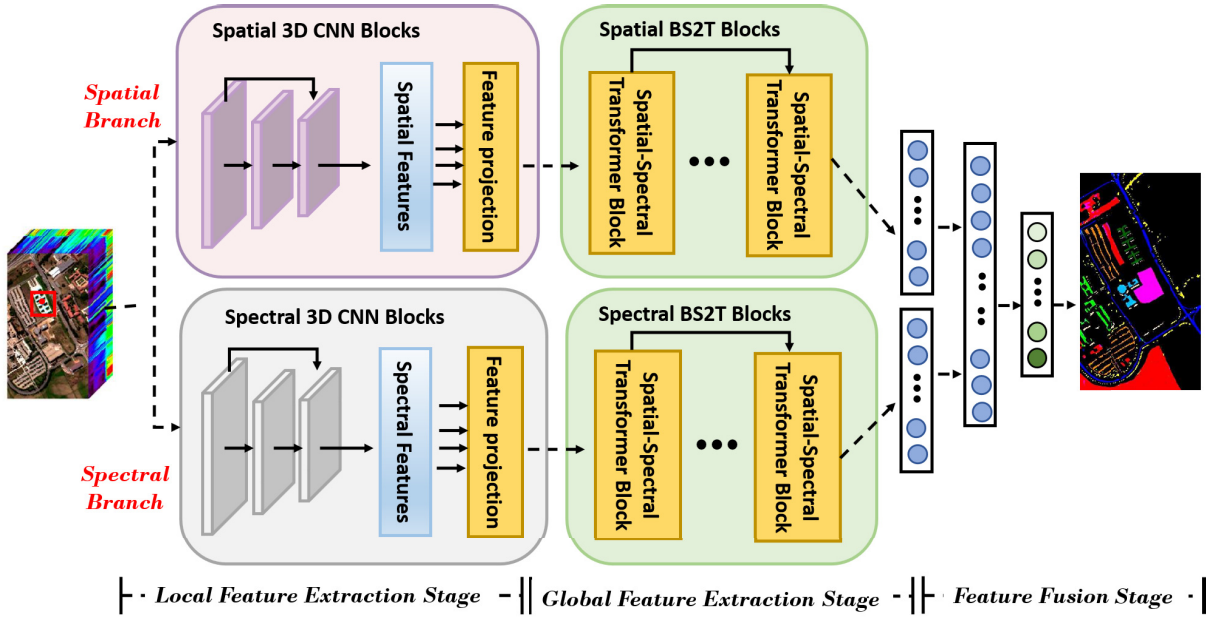


Fig. 2. Framework of the proposed BS2T HS classification network. The network consists of three stages. The first stage extracts the local feature of HS image pixels by 3-D convolutional operations. By defining the spatial–spectral transformer block, the second stage transfers the local information to obtain the global information of HS image pixels. The features obtained from the spectral branch and spatial branch are fused in the last stage.

operation is adopted to generate the output feature with the size of $H \times W \times C_{out}$. In the feature expansion module, Conv Operation with 1×1 kernel size is adopted to make the output feature can adjust the network structure. Based on MHSA used in [36], by introducing the spectral information of HS image patches, we propose the MHS2A to make better use of the spectral information.

1) Relative Positional Encoding for HS Image Feature: In general transformer-based architectures, positional encoding is usually used as constraints to make the attention more sensitive to positions. For CV tasks, the relative distance-aware positional encoding has proven its potential in depicting image-to-image spatial content positions. This can be attributed to attention not only considering the content information but also relative distances between features at different locations, thereby being able to effectively associate information across objects with positional awareness.

In our BS2T, we adopt the 2-D relative position self-attention implementation from [52]. The 2-D relative height information R_h and relative width information R_w are calculated for each HS image feature, where i and j represent the relative position of the HS feature, $F = \{f(i, j) \in \mathbb{R}^{H \times W \times C}\}$, and H, W, and C are the height, width, and number of channels of the feature, respectively.

2) Multihead Spatial–Spectral Self-Attention: The architecture of the proposed MHS2A is shown in Fig. 4. The original MHSA can be described as mapping of a query and a set of key–value pairs to the output, where the query, key, value, and output are all vectors. The output attention is calculated as a weighted sum of values, where the weights assigned to each value are calculated by the query's compatibility function with the corresponding key. In order to make the attention operation more position aware and spectral aware, the MHS2A with positional encoding is proposed to jointly depict the spatial–positional, spatial–spatial, and spatial–spectral correlations among the HS image patch.

HS image feature F is first fed into three convolution layers to obtain three sets of new features $\{Q, K, V\} \in \mathbb{R}^{H \times W \times C}$. At the same time, the global pooling operation maps the entire spatial feature on a channel as a global feature to obtain the spectral feature S of F

$$S_c = \frac{1}{H \times W} \sum_{i}^{H} \sum_{j}^{W} f(i, j), \quad S \in \mathbb{R}^{C}. \tag{2}$$

By introducing the spectral feature S, the proposed spatial–spectral attention mechanism is defined as

$$\text{Attention}(Q, K, V, S) = \text{softmax}\left(\frac{(R_h + R_w)QK^T}{\sqrt{C_k}}\right)SV \tag{3}$$

where R_h and R_w are the relative position encodings for height and width, respectively; Q, K, V, and S represent the query, key, value, and spectral, respectively; and C is the dimension of the input data. The proposed MHS2A with H heads can be then expressed as

$$\text{MHSA}(F) = \text{Concat}(h_1, h_2, \ldots, h_H)W^0 \tag{4}$$

where W^0 represents the learned parameters and h_i represent the attention obtained in each head

$$h_i = \text{Attention}\left(FW_i^Q, FW_i^K, FW_i^V, FW_i^S\right). \tag{5}$$

In the proposed MHS2A, different heads work independently in a parallel pattern to learn different attention. The attention for each head is calculated by (3) shown in Fig. 5 and all the attention is concatenated to obtain the output MHS2A.

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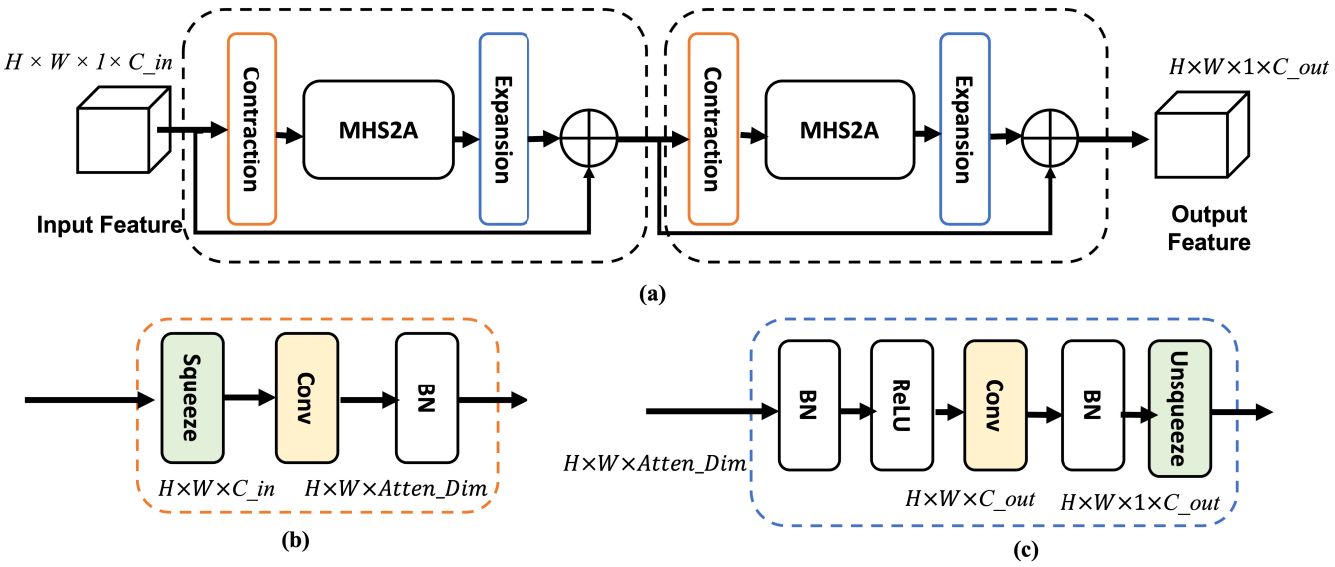


Fig. 3. Overall pipeline of the proposed HS image bottleneck spatial–spectral block. The figure is an example of a two-layer BS2T transformer block. (a) Overall pipeline. (b) Feature contraction module. (c) Feature expansion module.

Fig. 5(a) shows an illustration of the proposed MHS2A; in Fig. 5(a), we have three attention heads in total. Also, Fig. 5(b) shows how each head of MHSA works in an arbitrary region.

3) Analysis on the Proposed BS2T: The proposed BS2T is inspired by the bottleneck transformer [36] for computer vision. Different from the bottleneck transformer, we use the proposed MHS2A to depict the long-range dependences of HS image patches, which can effectively address the abundant spectral information of HS images.

From Fig. 4, it can be observed that, in MHS2A, three kinds of information relationships are depicted. First, the positional information is combined with Q to make the attention more position aware, where spatial–positional relation is addressed in the attention. Second, similar to dot-product attention, Q is combined with V to reflect the influence of spatial–spatial relations between different positions. Finally, the HS spectral information S is introduced to address the abundant spectral information and make the attention more spectral aware. The MHS2A can jointly depict the spatial–positional, spatial–spatial, and spatial–spectral correlations among the HS image patch, thus improving the network efficiency.

B. BS2T for HS Image Classification

By introducing the proposed HS-BS2T block, in this section, a novel dual-branch HS image classification framework based on 3-D CNN and BS2T is proposed for joint local–global feature extraction of HS images. As shown in Fig. 2, a spatial branch $\mathcal{G}_{\text{Spa}}$ and a spectral branch $\mathcal{G}_{\text{Spe}}$ consist in the proposed framework to simultaneously extracting the spatial–spectral features of the original HS image data. For each branch, the original HS data cubes are initially fed into dense 3-D CNN feature extraction backbones to obtain the global features. On this basis, the proposed BS2T blocks are used to capture the global long-range band and pixel dependencies of the HS data cubes. Finally, the obtained HS spatial–spectral features are fused to get the final classification result.

1) Network Structure: Assume that $\mathbf{X} = \{x_1, x_2, \ldots, x_N\}$ is the HS image dataset, where N is the number of pixels within the HS image cube and B represents the number of spectral bands. Its corresponding categorical label is defined as $\mathbf{Y} = \{y_1, y_2, \ldots, y_N\}$, where K is the land-cover categorical types contained in the area. To generate the initial training and testing data of the HS image $\mathbf{P} = \{p_1, p_2, \ldots, p_N\} \in \mathbb{R}^{r\times r\times B}$, for each HS image pixel, the corresponding HS patch p_i is the HS image patch centered at pixel x_i with a radius of r.

For the spatial branch $\mathcal{G}_{\text{Spa}}$, the HS image patches are first fed into the spatial 3-D CNN blocks to obtain HS image local spatial features F^L_{spa}. We adopt the similar network structure as in [27]. First, 3-D convolution with convolutional kernel of $3 \times 3 \times 1$ is adopted to extract the HS image spatial semantic information. For each spatial 3-D CNN block, only 3-D convolution, 3-D batch normalization, and Mish operations are included. For the dense connection between the 3-D CNN blocks, two neighboring dense blocks integrated together, but there is no overlap between every two blocks. Meanwhile, it is notable that, in the design of our network structure, to avoid overfitting and early stopping, we adopt the Mish operation [53] instead of Relu as the activation function. The formulation of Mish is defined as

$$\text{mish}(x) = x \times \tanh(\text{softplus}(x)) = x_i \times \tanh\left(\ln\left(1 + e^x\right)\right) \quad (6)$$

where x denotes the input network features.

The detailed implementation of the spatial 3-D CNN branch is shown in Fig. 6. The global spatial features F^G_{spa} and the local spatial feature F^L_{spa} obtained by the spatial 3-D CNN blocks are then fed into the spatial–spectral transform blocks. As illustrated in Section III-A, the adopted spatial–spectral transform block is constructed by the proposed HS spatial–spectral self-attention modules, thus enhancing the long-range dependency.

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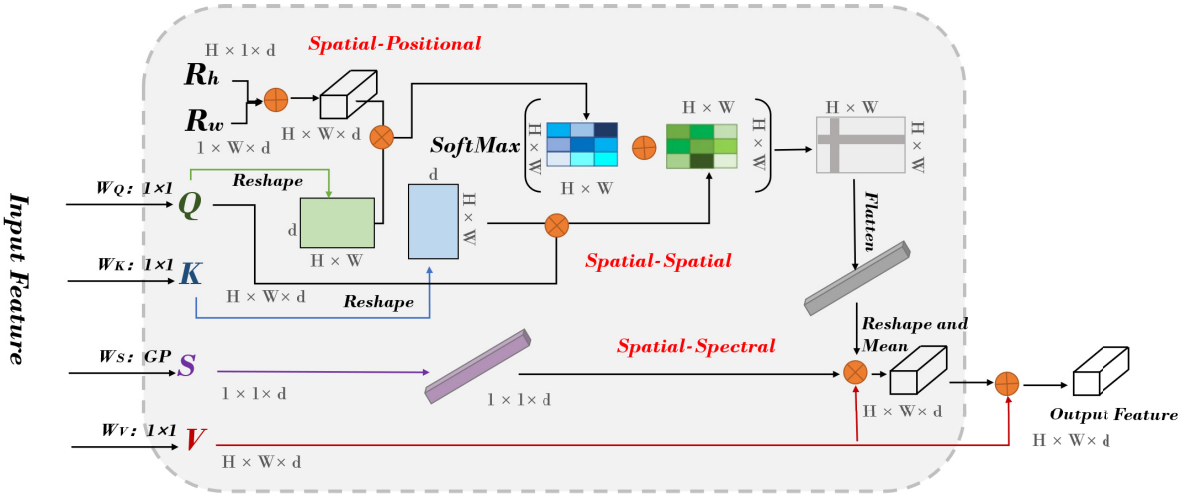


Fig. 4. Illustration of the single-head spatial–spectral self-attention, the input feature is transformed into Q, K, V, and S features to generate the attention maps. R_h and R_w are the positional encoding of the input features.

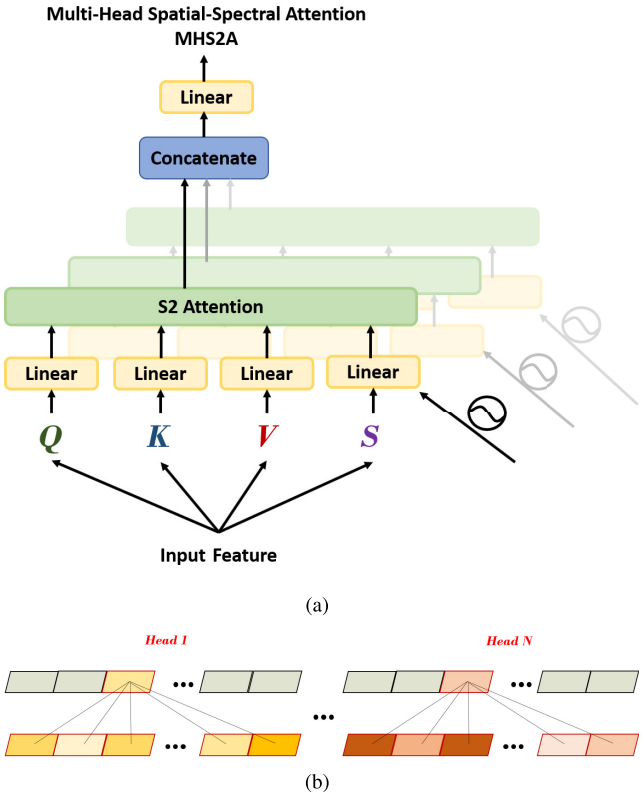


Fig. 5. Illustration of the proposed MHS2A. (a) MHS2A with three heads. (b) Visualization of the MHSA mechanism in a flattened arbitrary region. For each head, there is a learned attention distribution over the region.

For the spectral branch $\mathscr{G}_{\text{Spe}}$, a local information extraction stage and a global information extraction stage are, respectively, utilized to obtain the local spectral feature F^L_{spe} and global spectral feature F^G_{spe} of the initial HS image patches. Different from the spatial branch, in the spectral branch, we use 3-D convolution with conv kernel of $1 \times 1 \times 7$ to extract the HS image spectral information, and 3-D convolution, 3-D batch normalization, and Mish operations are combined to form the spectral 3-D CNN dense blocks. The detailed implementation of the spectral 3-D CNN branch is shown in Fig. 7. Similarly, the obtained local spectral features F^L_{spe} are then fed into the HS spatial–spectral transform blocks to get the global spectral feature F^G_{spe}. After obtaining F^L_{spe} and F^G_{spe}, to fuse the HS image spatial–spectral features, before fed into the fully connected layers, concatenation operation is adopted between F^L_{spe} and F^G_{spe}. The output of the fully connected layers can be depicted as

$$FC\left(F^L_{\text{spe}}, F^G_{\text{spe}}\right) = f\left(W \bullet \left(F^L_{\text{spe}} \middle|\middle| F^L_{\text{spa}}\right) + b\right) \tag{7}$$

where $||$ denotes the concatenation operation and W and b , respectively, represent the weight and bias of the fully connected layer. Then, the joint feature, denoted as F_M, is fed into the softmax classify layer to predict the probability distribution expressed as

$$\text{pred}(i, j) = \frac{1}{\sum_{j=1}^{K} \exp\left(\theta_j | F_M\right)} \begin{bmatrix} \exp\left(\theta_1 | F_M\right) \\ \cdots \\ \exp\left(\theta_K | F_M\right) \end{bmatrix} \tag{8}$$

where θ_j is the jth column of the weights in the prediction layer, K is the number of classes, and pred (i, j) is a 1-D vector, which decides the label of testing pixel.

2) Training Strategies: In the process of network training, the training set samples are fed into the proposed framework in the form of $\{p_i, y_i\}_{i=1}^{M}$, where p_i is the HS image training patches, y_i is the real labels corresponding to the pairs, and M is the total number of training samples. In order to deal with the problem of uneven quantities between the categories, we use the weighted cross-entropy function as the final loss function

$$\mathscr{L}_{W-C} = -\frac{1}{M} \sum_{i=1}^{M} \sum_{j=1}^{K} \omega_j \log \Pr(y_i = j | p_i, \theta) \tag{9}$$

where ω_j is the jth weight and $\Pr(y_i = j | p_i, \theta)$ is the probability that the pixel belongs to the jth class. For weights,

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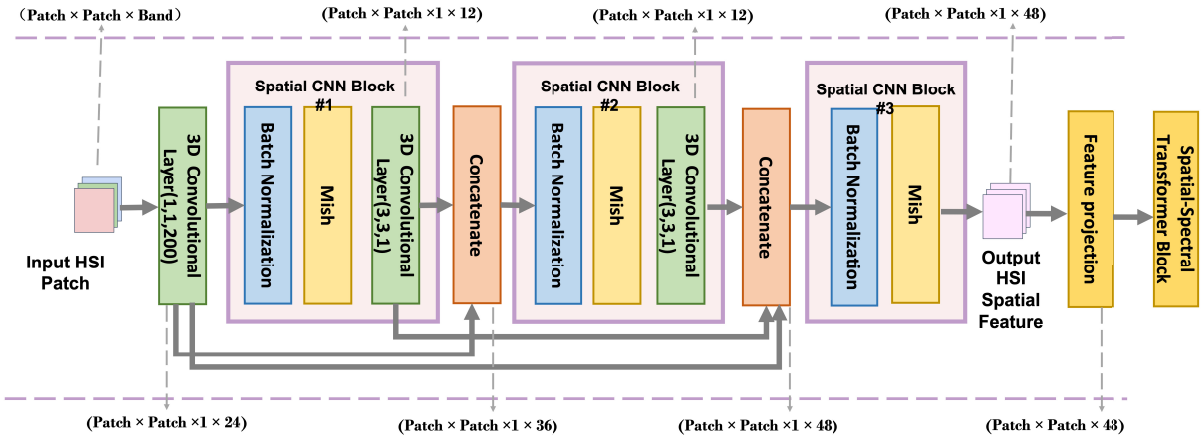


Fig. 6. Detailed implementation for the spatial 3-D CNN blocks. The output features of the #3 block are then projected to the BS2T block.

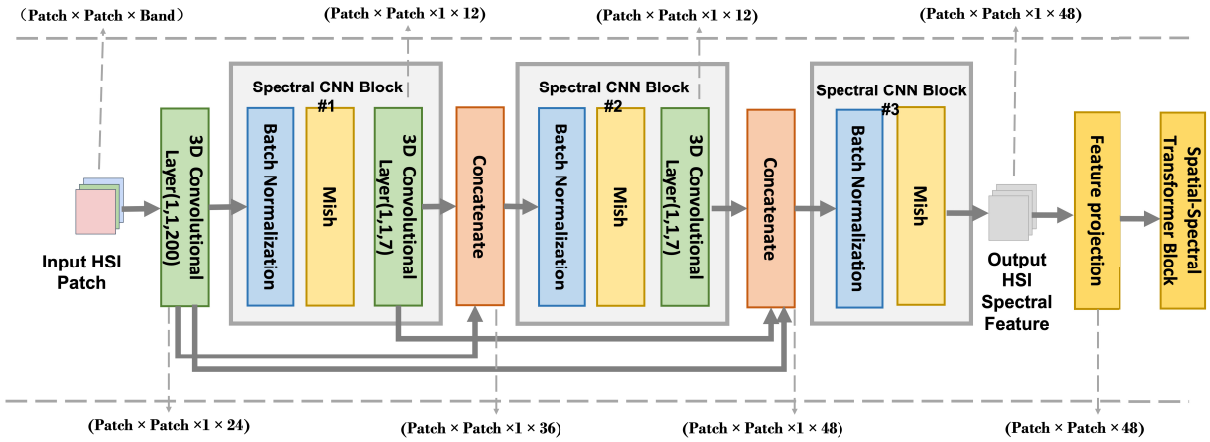


Fig. 7. Detailed implementation for the spectral 3-D CNN blocks. The output features of the #3 block are then projected to the BS2T block.

the definition is given as follows:

$$\omega_j = \frac{n_j}{M} \tag{10}$$

where n_j is the number of class j in the ground-truth training sample. Since the above function is continuously differentiable, we use the Adam optimizer with Nesterov momentum to obtain the maximum parameter [54].

IV. Experiments

To evaluate the effectiveness of the proposed HS image classification method, four public HS image classification datasets are used for experiments. In this section, we first present the detailed information of the used datasets. Then, we introduce the implementation details of the experiments. Finally, we compare the results of the datasets with some comparison algorithms and ablation experiments are conducted to demonstrate the effects of different modules.

A. Datasets Description

In this article, we use four public HS classification datasets to evaluate the effectiveness of the proposed BS2T net, including the Indiana Pines dataset (IN), Pavia University dataset (UP), Botswana dataset (BS), and Houston dataset (HU).

1) *Indiana Pines Dataset (IN):* The IN dataset was captured by the AVIRS HS sensor and depicts a farmland area in Northwestern Indiana. For the IN dataset, after the spectral, spatial, and atmospheric correction process, 200 spectral bands were selected for the experiments. The size of the selected region contains 145 × 145 pixels and 16 land-cover classes.
2) *Pavia University Dataset (UP):* The UP dataset was captured by ROSIS-3 HS sensor on the University of Pavia and its adjacent areas in 2001. For the UP dataset, after the spectral, spatial, and atmospheric correction process, 103 spectral bands were selected for the experiments. The size of the selected region contains 610 × 340 pixels and nine land-cover classes.
3) *Botswana Dataset (BS):* The BS dataset was captured by the EO-1 HS sensor on the Okavango Delta, CA, USA, in 2001. For the BS dataset, after the spectral, spatial, and atmospheric correction process, 145 spectral bands were selected for the experiments. The size of the selected region contains 1476 × 256 pixels and 14 land-cover classes.
4) *The Houston Dataset (HU):* The HU dataset is provided by the GRSS Data Fusion Contest 2013. The scene

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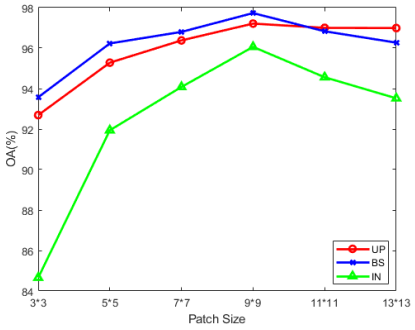


Fig. 8. Model performance comparison under different input patch sizes.

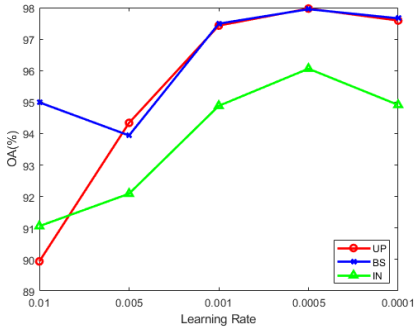


Fig. 9. Model performance comparison under different network learning rates.

was captured by airborne sensors on the University of Houston campus and its adjacent areas in 2012. For the HU dataset, after the spectral, spatial, and atmospheric correction process, 144 spectral bands were selected for the experiments. The size of the selected region contains 349 × 1905 pixels and 15 land-cover classes.

B. Parameter Tuning

The performance of a deep learning model is closely related to the design of network structure. In this section, we compare the classification results using different initial input data patch sizes and different learning rates with randomly selected training set.

1) Multiscale Input Patch Size Comparison: The input HS image patch size determines the information for each input, and the amount of information will have a certain impact on whether the BS2T modules can work effectively. In order to compare the influence of the input HS image patch size on the model, we compare the classification performance of the three datasets with different input patch sizes. Fig. 8 shows the classification performance of the three datasets with different input patch size ranging from 3 × 3 to 13 × 13. It can be observed from the results that the input patch sizes influence the performance significantly. The most suitable patch size is 9 × 9 for all the datasets.

2) Learning Rate Comparison: As one of the important superparameters in deep learning, the learning rate determines whether and when the objective function can converge to the local minimum value. Selecting the appropriate learning rate can make the objective function converge to the local minimum value in a limited time. Fig. 9 shows the classification performance of the proposed framework in three datasets with different learning rates. As can be seen from the results, the best learning rate for the three datasets is 0.0005.

C. Experimental Settings

The following conditions hold.

1) In the training sample generation process, two kinds of sampling strategies are considered in our experiments, including random sampling strategy and disjoint sampling strategy. In the experiments with randomly selected training samples, we use the IN dataset, UP dataset, and BS dataset as the testing datasets. Each dataset is divided into three subsets, including training set, validation set, and testing set. Deep learning-based HS image classification methods work in a data-driven way, and the amount of training sets directly influences the classification results. However, more training samples will bring extra computational cost. To make a balance between classification accuracy and computational efficiency, for the IN dataset, we first randomly select 201 pixels (about 1.96% of the total pixels) as the initial training set and 200 pixels as the validation set. As the samples are enough for the UP dataset and BS dataset, we, respectively, select 210 pixels and 40 pixels (about 0.5% of the total pixels) as the initial training set and 210 and 40 pixels as the validation set. In the experiments with spatially disjoint training samples, we use the disjoint IN dataset, disjoint UP dataset, and HU dataset as the testing datasets. Also, we adopt a sampling strategy based on spatially disjoint samples, and these spatially disjoint training and testing samples are available from the GRSS DASE website[1].
2) To verify the effectiveness of the proposed framework, some recent proposed classifiers are used as the comparison methods, including: SVM [10], contextual deep CNN (CDCNN) [55], dual-level deep spatial manifold representation (SMR-EG) [56], attention-based adaptive spectral–spatial kernel ResNet (A2S2K) [57], spectral–spatial residual network (SSRN) [31], double-branch multiattention mechanism network (DBMA) [41], DBDA [27], and ViT [39]. To make a fair comparison, the patch size is set as 9 × 9 for SMR-EG, A2S2K, SSRN, DBMA, DBDA, and the proposed method, and the patch size is set as 5 × 5 for CDCNN. Meanwhile, the learning rate is set as 0.0005 for all the deep learning methods.
3) The overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa) are selected as the evaluation metrics for the quantitative evaluation. All the involved models have been run ten times, and the collected average result is reported for each experiment.

[1] http://dase.grss-ieee.org

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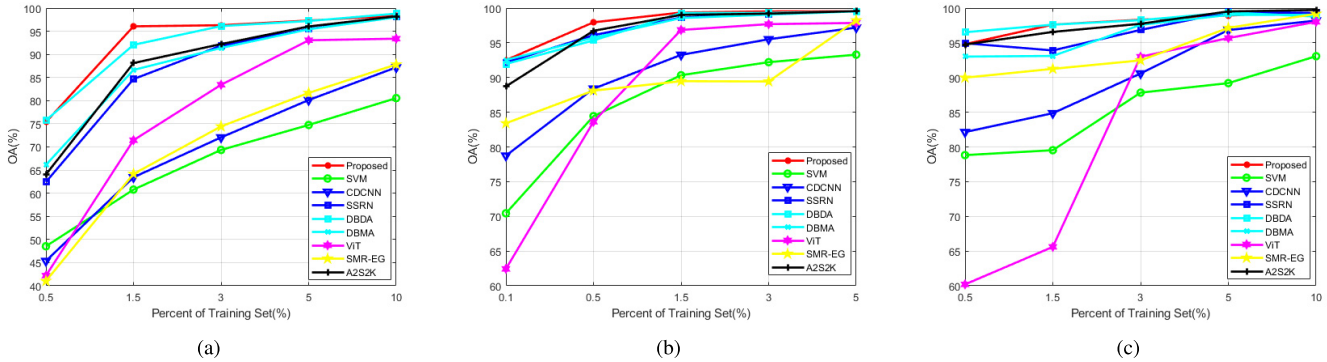


Fig. 10. Model performance comparison under different random selected training samples. (a) IN dataset. (b) UP dataset. (c) BS dataset.

4) For the implementation of our program, all the experiments are implemented using Python 3.6, and the CNNs are constructed with PyTorch. All experiments are conducted using a personal computer equipped with 6 GB of memory and an NVDIA GeForce RTX 1660.

D. Classification Performance Over Random Sampling Datasets

Tables I–III, respectively, give the quantitative classification results for the IN dataset, UP dataset, and BS dataset with randomly selected training sets. It can be observed from the tables that the proposed method outperforms the comparison methods, and a significant improvement can be observed. For example, in Table I, the proposed method reaches an OA of 94.29% for the IN datasets, and compared with the ViT method, the proposed method yields an 18.80% average absolute improvement in terms of OA and 21.52 absolute improvement in terms of Kappa, which demonstrates the effectiveness of the joint local–global feature extraction framework. Meanwhile, for classes with limited samples, such as the 1st, the 9th, and the 16th class of IN dataset, the improvement is more obvious; especially for the first class, only three samples are used for training and the accuracy reaches 97.29% in the testing set, which demonstrates the effectiveness of the proposed method under limited training samples. In summary, the proposed classification framework is able to consistently outperform the SOTA classification methods with the randomly selected training dataset.

Classification performance over different numbers of training samples is also conducted. Fig. 10 reports the classification results of different methods by varying the training samples. For the IN dataset and BS dataset, the sampling rate varies from 0.5% to 10%, and for the UP dataset, the sampling rate varies from 0.1% to 5%. As shown in Fig. 10, when the percentage is small, the proposed method shows a significant improvement upon other comparison methods. For example, in the IN dataset, when the sampling rate is set as 1.5%, the proposed method can achieve an OA of 96%, while for other deep learning-based method, such as ViT, the OA is just nearly 71%. The improvement gap is obvious. When the sampling rate is large, our method is still competitive among all the methods; however, the improvement is not such significant.

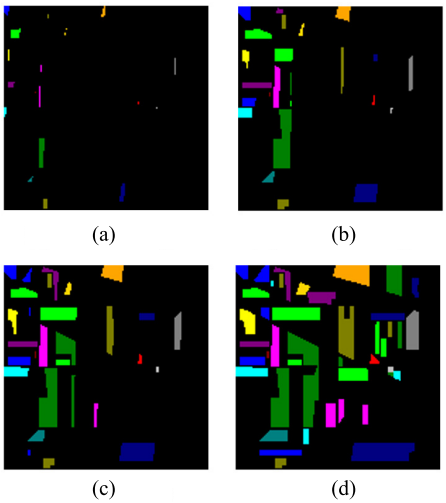


Fig. 11. Spatially disjoint training samples for the disjoint IN dataset. (a) 5% training. (b) 30% training. (c) 50% training. (d) IN disjoint training.

Overall, the proposed method shows obvious improvement in the general trend, and the improvement is more significant under limited training samples.

E. Classification Performance Over Disjoint Datasets

In the random sampling datasets, when extracting neighborhood patches for the training set, these patches may overlap the test information, to avoid the spatially overlapping phenomenon, we also conduct our experiments on the disjoint datasets, including disjoint IN dataset, disjoint UP dataset, and Houston dataset.

To demonstrate the generalization ability of our proposed model, we conduct the experiments over different numbers of disjoint training samples. For the disjoint IN dataset, we select 5%, 30%, and 50% samples of the IN disjoint training set. For the disjoint UP dataset, we select 8%, 30%, and 50% samples of the UP disjoint training set. Similarly, for the HU dataset, with the most limited training set, we select 10%, 30%, and 50% samples of the original training set. The training samples of each dataset are shown in Figs. 11–13.

Table IV shows the classification results with different training percentages over disjoint IN dataset. It can be observed

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TABLE I

CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR THE IN DATASET (201 LABELED TRAINING SAMPLES)

Class.	SVM-Ref	CDCNN	SMR-EG	A2S2K	SSRN	DBMA	DBDA	ViT	Proposed
1 Alfalfa(1/44)	31.95±0.00	58.03 11.41	33.14±18.90	96.01±5.31	81.07±13.35	69.56±10.88	96.81±2.25	54.07±22.60	**97.29±2.60**
2 Corn-notill(28/1372)	52.88±0.00	59.69±3.68	53.64±4.75	82.76±3.21	86.85±2.07	81.75±370	88.35±2.09	68.95±9.76	**90.46±3.77**
3 Corn-mintill(16/798)	55.66±0.00	52.34±9.14	42.35±4.00	85.92±1.58	87.90±2.84	86.41±3.06	**91.14±2.33**	66.12±12.03	89.62±3.45
4 Corn(5/227)	36.73±0.00	63.87±9.14	47.24±10.33	95.37±2.75	88.88±3.30	88.35±5.79	**97.61±2.38**	62.25±16.08	97.48±3.67
5 Grass-pasture(10/463)	82.53±0.00	91.72±4.46	75.47±6.16	98.30±1.97	**98.89±0.89**	97.57±1.09	98.36±0.61	83.95±21.81	98.40±0.24
6 Grass-trees(14/702)	80.66±0.00	85.40±3.10	73.33±9.82	93.86±2.94	95.04±1.53	87.68±2.36	96.11±1.59	88.55±7.80	**96.76±0.94**
7 Grass-pasture-mowed(1/26)	44.97±0.00	42.15±10.23	19.33±4.97	57.77±4.04	**68.70±7.08**	39.84±10.80	43.42±1.02	29.42±13.98	44.30±4.33
8 Hay-windrowed(9/460)	89.09±0.00	91.76±2.10	86.90±5.93	99.13±0.78	99.22±0.83	99.21±0.45	99.95±0.08	95.37±3.20	**100.00±0.00**
9 Oats(1/18)	34.99±0.00	54.94±18.45	26.76±17.90	51.54±5.82	66.00±8.91	31.73±13.32	**59.92±9.03**	35.23±19.92	55.76±14.91
10 Soybean-notill(19/934)	57.19±0.00	64.88±4.75	58.21±10.58	88.73±3.80	90.48±3.38	85.43±1.25	**92.41±1.27**	72.72±12.05	91.30±1.22
11 Soybean-mintill(48/2359)	57.74±0.00	62.98±1.30	66.51±4.51	86.16±1.58	90.11±1.76	90.34±2.00	94.56±1.71	80.79±11.11	**95.49±2.10**
12 Soybean-clean(11/571)	39.59±0.00	37.84±6.09	51.25±3.39	88.11±2.34	88.83±5.13	86.06±4.01	98.76±0.55	72.83±18.62	**98.90±0.79**
13 Wheat(4/197)	84.86±0.00	79.98±7.80	58.86±18.52	98.41±2.20	97.78±0.68	**100.00±0.00**	98.88±1.21	86.74±17.26	99.48±0.79
14 Woods(25/1215)	84.89±0.00	83.24±3.93	86.20±3.89	95.08±1.48	95.12±1.36	95.83±2.38	**98.37±0.73**	89.31±8.62	97.92±0.60
15 Building-Grass-Trees-Drives(7/372)	62.24±0.00	67.90±6.48	54.81±13.82	**93.81±2.59**	88.67±4.09	67.87±5.77	92.77±1.91	74.27±12.55	92.36±2.66
16 Stone-Steel-Towers(2/89)	**98.53±0.00**	84.09±10.05	71.75±9.48	93.92±2.38	89.75±3.19	83.60±4.91	92.74±2.43	84.45±13.03	92.66±3.06
OA(%)	63.89±0.00	67.59±1.10	62.70±2.57	89.13±0.87	90.96±0.54	87.40±0.86	94.08±0.37	75.49±7.20	**94.29±0.56**
AA(%)	62.16±0.00	67.55±2.28	56.61±3.21	87.81±0.80	88.33±0.64	80.70±1.76	90.04±0.89	71.56±5.81	**91.26±1.13**
Kappa	58.09±0.00	62.67±1.25	57.29±2.98	87.54±1.02	89.68±0.62	85.63±0.98	93.24±0.43	71.96±8.46	**93.49±0.64**

TABLE II

CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR THE UP DATASET (210 LABELED TRAINING SAMPLES)

Class.	SVM-Ref	CDCNN	SMR-EG	A2S2K	SSRN	DBMA	DBDA	ViT	Proposed
1 sphalt(33/6573)	90.23±0.00	86.96±2.29	78.10±3.42	92.18±2.58	95.60±0.63	89.15±0.83	94.60±1.43	57.60±38.68	**95.66±2.75**
2 Meadows(93/18474)	86.41±0.00	93.38±2.26	97.18±0.84	97.18±0.84	97.57±1.14	97.97±0.91	**99.09±0.52**	63.52±41.93	98.63±0.59
3 Gravel(10/2079)	52.45±0.00	68.46±8.65	55.08±8.00	93.68±3.79	91.16±5.49	87.01±5.94	**95.02±2.78**	16.86±19.22	92.97±1.85
4 Trees(15/3029)	85.68±0.00	97.62±2.47	82.40±7.32	97.97±3.00	99.61±0.42	96.96±0.98	96.76±1.78	61.10±41.40	**98.50±0.57**
5 Painted metal sheets(6/1332)	99.74±0.00	98.96±1.01	87.95±5.66	**99.62±0.34**	99.80±0.16	99.28±0.30	98.65±0.81	62.49±41.66	99.07±0.33
6 Bare Soil(25/4966)	63.46±0.00	86.55±1.00	87.57±3.14	96.68±1.67	97.46±1.52	96.27±1.29	98.04±0.46	40.80±32.27	**97.78±0.99**
7 Bitumen(6/1318)	64.84±0.00	76.37±7.28	62.47±15.79	80.72±15.59	83.68±10.84	93.38±5.36	97.20±2.74	24.93±34.97	**94.62±7.14**
8 Self-Blocking Bricks(18/3649)	73.32±0.00	75.24±4.72	55.52±4.67	88.17±2.96	87.24±3.68	87.75±1.66	89.82±3.60	47.10±32.05	**92.07±2.41**
9 Shadows(4/936)	**100.00±0.00**	97.67±0.94	57.95±10.55	97.64±1.92	97.56±1.68	97.23±2.03	97.15±1.13	64.51±42.86	98.22±0.98
OA(%)	80.96±0.00	88.14±0.67	82.38±1.61	94.84±0.78	95.66±0.83	94.73±0.24	96.92±0.37	79.49±6.76	**97.03±0.69**
AA(%)	79.57±0.00	86.62±1.27	73.38±3.57	93.83±1.27	94.43±1.55	93.89±1.36	96.26±0.18	69.67±11.08	**96.39±0.93**
Kappa	74.47±0.00	84.08±0.96	76.46±2.19	93.13±1.06	94.23±1.12	92.98±0.32	95.92±0.50	72.66±8.99	**96.06±0.93**

TABLE III

CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR THE BS DATASET (42 LABELED TRAINING SAMPLES)

Class.	SVM-Ref	CDCNN	SMR-EG	A2S2K	SSRN	DBMA	DBDA	ViT	Proposed
1 Water(3/265)	100.00±0.00	93.14±4.89	76.54±13.31	98.13±1.46	99.11±0.94	**99.17±0.46**	98.59±0.62	90.84±12.55	99.03±0.71
2 Hippo grass(2/97)	87.87±0.00	64.58±16.64	54.80±17.20	83.93±15.15	86.28±10.04	**99.79±0.60**	99.21±1.72	68.05±29.41	94.42±5.52
3 Floodplain grasses1(3/245)	76.64±0.00	73.24±6.58	70.58±21.26	98.79±2.88	97.86±2.98	99.94±0.15	99.68±0.96	95.58±6.15	**100.00±0.00**
4 Floodplain grasses2(3/209)	67.19±0.00	75.00±13.14	64.54±16.18	91.99±4.22	91.32±5.12	89.56±2.78	87.69±1.04	**93.32±7.19**	87.61±2.58
5 Reeds1(3/263)	88.11±0.00	78.49±5.22	60.26±13.88	**88.28±5.28**	89.42±3.63	89.27±1.91	77.65±4.14	79.93±17.91	81.35±5.66
6 Riparian(3/263)	57.26±0.00	73.45±7.52	66.13±13.11	90.42±8.35	91.25±9.45	97.30±4.67	99.88±0.33	78.07±27.99	**99.91±0.26**
7 Fierscar2(3/253)	92.10±0.00	97.88±2.43	94.99±6.36	96.81±2.66	98.89±1.60	94.36±4.61	**100.00±0.00**	97.56±3.66	99.88±0.18
8 Island interior(3/197)	83.06±0.00	80.78±4.98	63.98±17.31	89.78±7.38	91.59±4.64	99.59±0.84	98.99±1.79	88.21±14.52	**99.59±1.05**
9 Acacia woodlands(4/306)	77.72±0.00	90.83±7.89	76.56±13.58	91.43±5.19	91.66±4.63	**99.90±0.29**	97.75±2.66	88.79±12.50	99.51±0.93
10 Acacia shrublands(3/242)	59.27±0.00	88.67±4.72	69.15±7.80	92.06±5.22	92.06±5.26	87.23±2.83	95.54±1.92	81.02±10.22	**97.70±2.16**
11 Acacia grassland(4/297)	89.63±0.00	97.78±2.13	82.44±13.30	99.81±0.23	98.70±1.55	98.49±0.69	99.42±4.89	94.91±9.95	**99.49±0.71**
12 Short mopane(2/177)	72.93±0.00	83.49±10.36	93.31±7.29	96.69±5.77	99.03±1.00	98.81±2.25	99.45±1.63	87.01±19.22	**99.54±0.65**
13 Mixed mopane(3/262)	94.67±0.00	89.28±3.56	83.63±9.06	93.46±5.91	95.40±4.84	99.29±1.15	99.69±0.44	87.48±16.06	**99.77±0.56**
14 Exposed soils(1/93)	79.77±0.00	86.81±13.17	85.14±12.60	98.31±3.95	99.75±0.73	**100.00±0.00**	99.63±1.08	82.15±33.81	99.64±1.05
OA(%)	78.50±0.00	83.45±2.71	72.88±8.66	93.19±1.57	94.16±1.82	96.08±0.75	96.02±0.78	84.96±6.85	**96.73±0.51**
AA(%)	80.44±0.00	83.81±3.29	74.43±8.71	93.56±1.49	94.45±1.64	96.62±0.62	96.67±0.52	86.64±6.90	**96.96±0.57**
Kappa	76.73±0.00	82.09±2.92	70.65±9.33	92.62±1.70	93.67±1.97	95.76±0.81	95.68±0.85	83.72±7.40	**96.46±0.55**

from Table IV that the proposed method utterly outperforms the other comparison methods, especially when the training sample is extremely limited. For example, when the training percentage is set as 5%, only 200 disjoint sampling pixels are selected to train the network, and the proposed method can achieve an average OA result of 58.64%, which yields an average of 6.72% absolute improvement compared with the DBDA method and an average of 21.27% absolute improvement compared with the ViT method. When the training percentage is set as 100%, the proposed method still presents a competitive performance among all the methods.

Table V shows the classification results with different training percentages over disjoint UP dataset. It can be observed from Table IV that for the disjoint UP dataset, all

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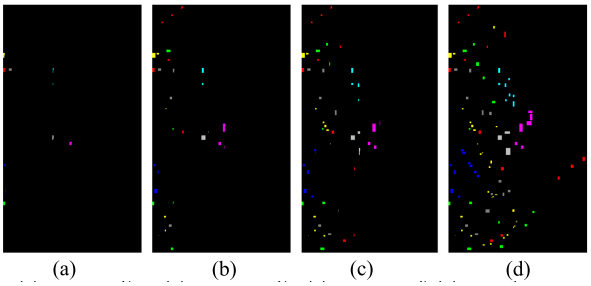


Fig. 12. Spatially disjoint training samples for the disjoint UP dataset. (a) 8% training. (b) 30% training. (c) 50% training. (d) UP disjoint training.

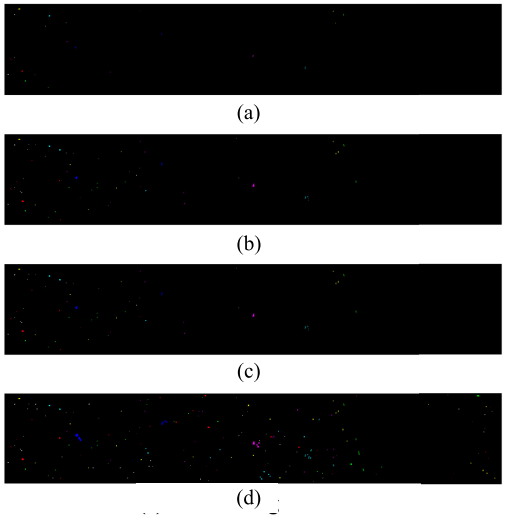


Fig. 13. Spatially disjoint training samples for the HU dataset. (a) 10% training. (b) 30% training. (c) 50% training. (d) HU training.

TABLE IV
CLASSIFICATION RESULTS WITH DIFFERENT TRAINING PERCENTAGES OVER DISJOINT IN DATASET

Disjoint IN	5%	30%	50%	100%
CDCNN	34.08±1.54	56.55±1.60	65.35±2.88	80.45±2.26
SMR-EG	43.09±4.00	50.89±3.42	53.22±3.30	68.45±5.19
A2S2K	22.60±4.26	54.37±5.37	68.87±4.39	78.43±10.66
SSRN	44.78±6.47	57.75±3.47	68.51±5.51	87.55±3.05
DBMA	49.60±4.57	57.59±3.97	70.13±4.92	87.74±3.02
DBDA	51.62±3.69	60.28±1.82	74.51±2.52	90.13±1.11
ViT	37.07±2.61	57.49±3.87	66.98±3.35	84.61±3.84
Proposed	**58.34±3.33**	**70.49±4.32**	**78.01±1.04**	**91.09±0.84**

the comparison methods present satisfaction results, and the results of the proposed method present a smaller standard deviation, which indicates the robustness of our proposed method. Table VI shows the classification results with different training percentages over Houston dataset. Similar to the disjoint IN dataset, the proposed method presents a significant improvement against the other comparison methods, especially when the training sample is limited. Among the comparison methods, when the training sample is limited, SMR-EG presents an extremely poor performance, and this is because SMR-EG utilizes nonlocal information of the training samples; when the training sample is extremely scattered, the accuracy is thus limited.

TABLE V
CLASSIFICATION RESULTS WITH DIFFERENT TRAINING PERCENTAGES OVER DISJOINT UP DATASET

Disjoint UP	8%	30%	50%	100%
CDCNN	39.80±0.88	68.45±1.47	**74.12±1.66**	84.97±2.13
SMR-EG	46.00±10.83	64.95±8.38	63.01±7.08	79.10±621
A2S2K	51.08±2.99	63.62±5.64	64.58±7.29	87.45±2.79
SSRN	49.88±6.42	66.72±6.32	64.97±3.92	84.43±2.65
DBMA	51.01±5.88	**70.20±1.98**	66.58±3.79	87.63±1.83
DBDA	46.18±3.48	68.90±0.69	64.17±3.53	84.42±2.39
ViT	**52.62±7.67**	65.00±3.84	66.95±2.16	79.49±6.76
Proposed	49.70±2.79	64.52±1.02	66.61±0.96	**87.81±0.81**

TABLE VI
CLASSIFICATION RESULTS BY DIFFERENT METHODS WITH DIFFERENT TRAINING PERCENTAGES OVER DISJOINT HU DATASET

Houston	10%	30%	50%	100%
CDCNN	50.74±1.36	58.44±0.91	72.05±1.18	78.93±1.44
SMR-EG	8.60±1.51	7.60±2.35	63.89±5.74	70.21±7.42
A2S2K	48.74±12.62	44.22±14.82	74.95±2.28	85.41±1.46
SSRN	42.89±12.06	47.19±16.09	73.78±1.46	86.55±2.36
DBMA	50.73±4.36	51.85±7.88	74.37±2.35	84.03±1.31
DBDA	48.10±10.45	58.55±2.84	76.78±2.05	86.75±0.62
ViT	29.02±11.85	51.23±6.14	71.06±2.37	79.08±3.08
Proposed	**62.09±1.25**	**62.50±0.83**	**77.64±0.48**	**88.22±1.22**

Fig. 14 shows the classwise classification performance when the training sample is set as 100% using the CDCNN, ViT, and proposed method. From the results, for most of the classes, the proposed method achieves the best classification accuracy.

Visualization results of the SOTA methods on the IN dataset, UP dataset, and BS dataset as, respectively, presented in Figs. 15–17. It is easy to observe that the proposed method produces a better classification performance with clearer boundaries and less noise, which is consistent with the quantitative results reported in the above tables. For example, in Fig. 15, by the comparison of deep learning-based methods, such as CDCNN SSRN and SMR-EG, large amount of misclassification points with strong noise can be observed, especially in small-sized regions, while the proposed method produces less misclassification points, even for the small-sized regions; this is because of the introduction of the BS2T module, which depicts the long-range global dependency of HS images, and thus, the class with limited samples has been greatly enhanced.

F. Model Complexity Analysis

In this section, the complexity of the proposed method and other comparison methods is studied in terms of running time, parameter size, and Flops of the networks. The model complexity of each method is reported in Table VII, where the training time is calculated using randomly selected training samples. From Table VII, it can be observed that, compared with other advanced deep learning methods such as SMG-EG, DBDA, DBMA, and ViT, the proposed method can achieve a better classification result with less network parameters and Flops. For instance, the proposed method only has 202 515 parameters of the UP dataset, which is nearly 20 times slighter compared with the ViT method.

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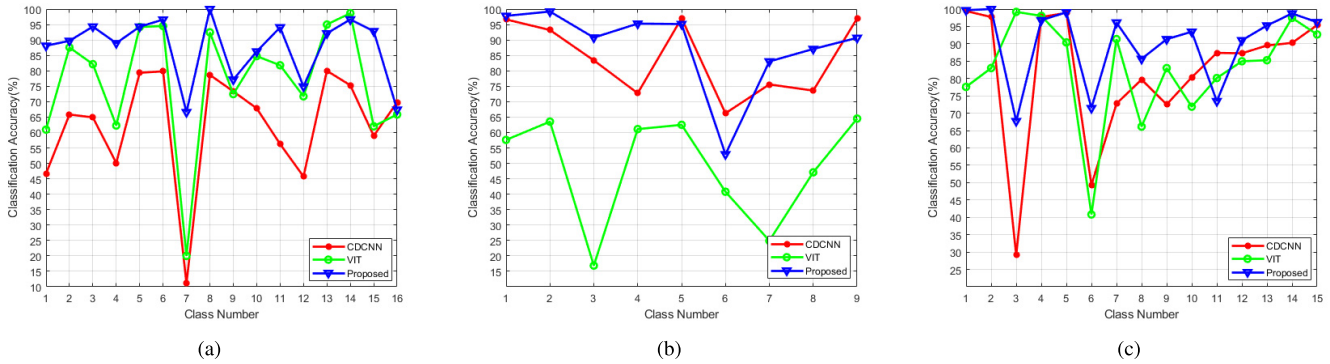


Fig. 14. Classwise classification accuracy. (a) Disjoint IN. (b) Disjoint UP. (c) HU.

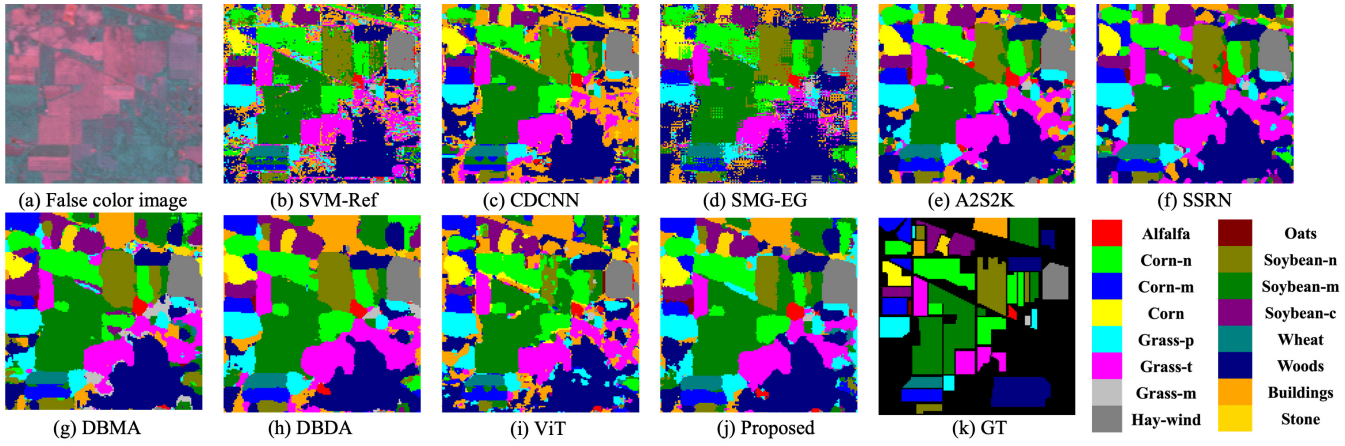


Fig. 15. Visualization of classification maps generated by different methods on the disjoint IN dataset. (a) False-color image. (b) SVM-Ref. (c) CDCNN. (d) SMG-EG. (e) A2S2K. (f) SSRN. (g) DBMA. (h) DBDA. (i) ViT. (j) Proposed. (k) GT.

It is notable that the training time of the proposed method is slightly slower than the comparison methods, and this is because the calculation of the spectral information of the M2SA module will bring extra computation time. However, overall, the training time of the proposed method is within a tolerable scope, which can achieve a balance between classification accuracy and computation efficiency.

G. Ablation Study

1) Ablation Study on Model Depth of BS2T: The model depth of BS2T refers to the number of spatial–spectral self-attention modules applied in each BS2T block. The number of spatial–spectral self-attention modules in each BS2T block influences the classification accuracy and computation efficiency. As illustrated in Section III, the proposed spatial–spectral attention module can depict the long-range dependency of HS image pixels; however, when excessive layers are adopted, the model dependency will become extremely weak and will lead to the network overfitting. To find the balance between classification accuracy with computation efficiency and avoid the overfitting phenomenon, by fixing all the other parameters, we vary the depth from 1 to 5 on the three used randomly selected HS classification datasets. The OA, AA, and Kappa of the three datasets with different model depths are reported in Table VIII. From the statistical results, it can be observed that, when the depth is 3, all the datasets can reach their best accuracy.

TABLE VII
MODEL COMPLEXITY COMPARISON

		training time /epoch(s)	Parameters	Flops (GMac)
IN	CDCNN	0.35	1,063,568	0.01
	SMG-EG	0.74	850,192	0.03
	A2S2K	1.46	369,534	0.17
	SSRN	1.19	364,168	0.16
	DBMA	1.52	609,791	0.24
	DBDA	1.51	606,906	0.24
	ViT	0.67	4,296,680	0.39
	Proposed	2.31	377,020	0.11
UP	CDCNN	0.35	628,105	0.01
	SMG-EG	0.79	586,249	0.02
	A2S2K	1.04	220,078	0.09
	SSRN	0.79	216,537	0.08
	DBMA	0.89	324,376	0.15
	DBDA	1.01	321,491	0.15
	ViT	1.51	4,147,425	0.3
	Proposed	2.51	202,515	0.05
BS	CDCNN	0.09	816,910	0.01
	SMG-EG	0.27	587,534	0.02
	A2S2K	0.37	285,518	0.12
	SSRN	0.28	281,174	0.11
	DBMA	0.34	446,949	0.21
	DBDA	0.33	444,064	0.21
	ViT	0.27	4,228,839	0.35
	Proposed	0.59	279,368	0.08

2) Ablation Study on Attentional Head Number in MHS2A: The proposed MHS2A can produce multiple attention maps and spatial–spectral feature embeddings of the HS image to encode abundant information, which enhances the robustness

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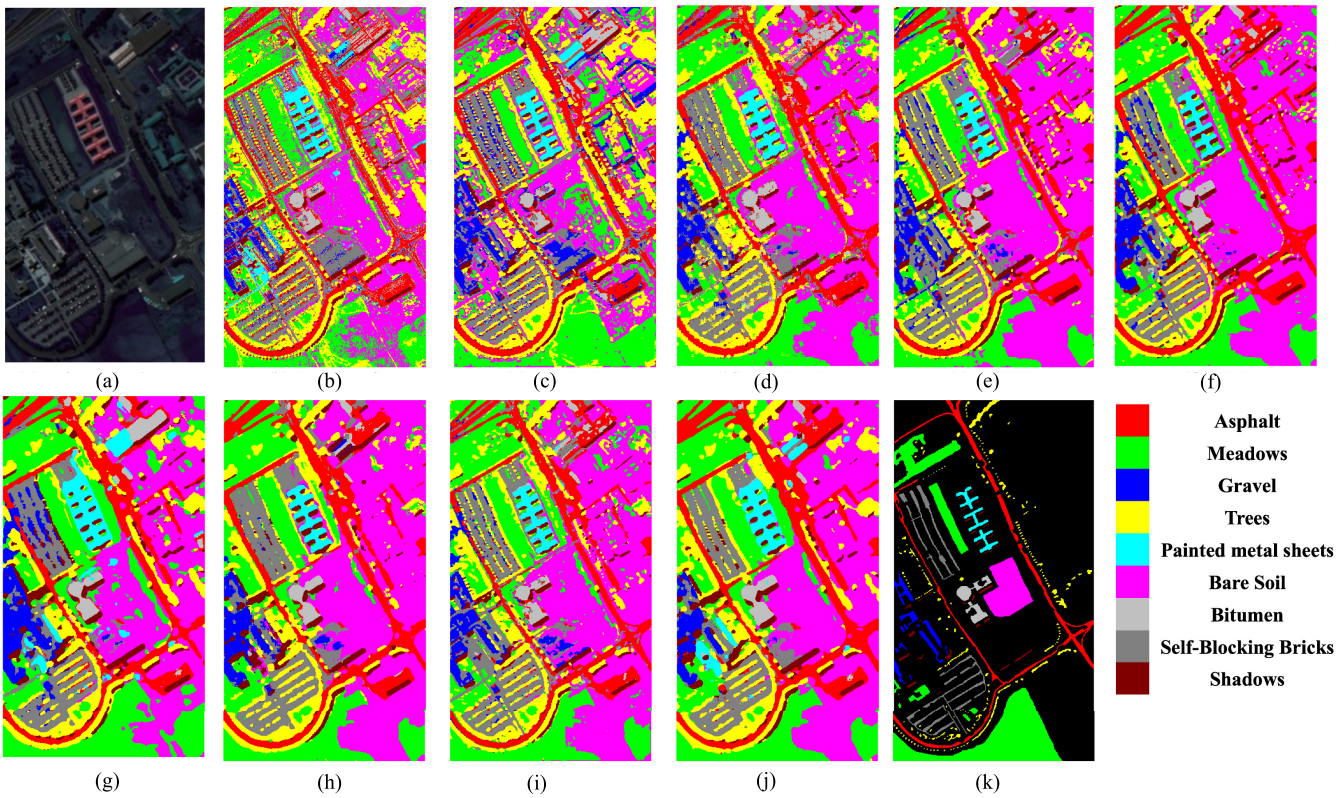


Fig. 16. Visualization of classification maps generated by different methods on the disjoint UP dataset. (a) False-color image. (b) SVM-Ref. (c) CDCNN. (d) SMG-EG. (e) A2S2K. (f) SSRN. (g) DBMA. (h) DBDA. (i) ViT. (j) Proposed. (k) GT.

TABLE VIII
ABLATION STUDY ON MODEL DEPTH OF BS2T

Layers		1	2	3	4	5
OA	IN	93.13±0.55	93.09±0.91	**94.29±0.56**	93.23±0.69	92.76±0.64
	UP	96.31±0.99	96.65±0.06	**97.03±0.69**	96.53±0.58	96.31±0.82
	BS	95.37±0.87	95.87±1.24	**96.73±0.51**	95.16±0.53	95.68±0.54
AA	IN	90.90±1.16	90.44±1.27	**91.26±1.13**	90.72±0.98	90.95±0.45
	UP	96.02±0.59	96.30±0.46	**96.39±0.93**	96.60±0.33	96.33±0.81
	BS	96.02±0.51	96.39±1.01	**96.96±0.57**	95.74±0.60	96.28±0.51
Kappa	IN	92.15±0.64	92.11±1.04	**93.49±0.64**	92.27±0.81	91.74±0.73
	UP	95.09±1.33	95.55±0.08	**96.40±0.93**	95.38±0.78	95.09±1.09
	BS	94.98±0.95	95.43±1.34	**96.46±0.55**	94.75±0.57	95.31±0.59

TABLE IX
ABLATION STUDY ON ATTENTIONAL HEAD NUMBER OF THE M2HA

Number of Heads		2	3	4
OA	IN	93.43±0.42	**94.29±0.56**	92.78±0.84
	UP	96.35±0.48	**97.03±0.69**	96.67±0.53
	BS	95.34±0.53	**96.73±0.51**	95.56±0.98
AA	IN	89.89±1.11	**91.26±1.13**	90.69±1.18
	UP	96.32±0.66	96.40±0.93	**96.79±0.32**
	BS	96.03±0.68	**96.96±0.57**	96.36±0.58
Kappa	IN	92.50±0.49	**93.49±0.64**	91.76±0.96
	UP	95.15±0.65	**96.40±0.93**	95.57±0.71
	BS	94.95±0.58	**96.46±0.55**	95.19±1.05

of the designed HSI classification network. The number of the selected attentional head number in MHS2A will influence the classification accuracy of the network. In MH2SA, different heads work independently in a parallel pattern to learn different the HS image features; when the selected head number is too large, they are tending to redundantly encode similar or the same information of the HS images and thus to influence the accuracy and efficiency of the network. In this section, by fixing all the other parameters, we vary the attentional head number in MHS2A from 2 to 4 on the three used randomly selected HS classification datasets. The OA, AA, and Kappa of the three datasets with different attentional head numbers are reported in Table IX. From the statistical results, it can be observed that, when the attentional head number is set as 3, all the datasets can reach their best accuracy.

3) Ablation Study on Attentional Modules: The attentional modules used in the BS2T play a significant role in the quality of the classification accuracy. In the proposed framework, we combine the traditional self-attention modules with positional information and spectral information. To demonstrate the effectiveness of the positional encoding and spectral information on the attention module, we fix all the other parameters to compare the classification results when only using self-attention, self-attention with positional encoding, and the proposed method. Table X shows the classification results on the three randomly selected training datasets, where SA represents only using self-attention, SA + PE represents using self-attention and positional encoding, and SA + PE + SI represents self-attention + positional encoding + spectral information. From the results, significant improvement can be observed by applying positional encoding

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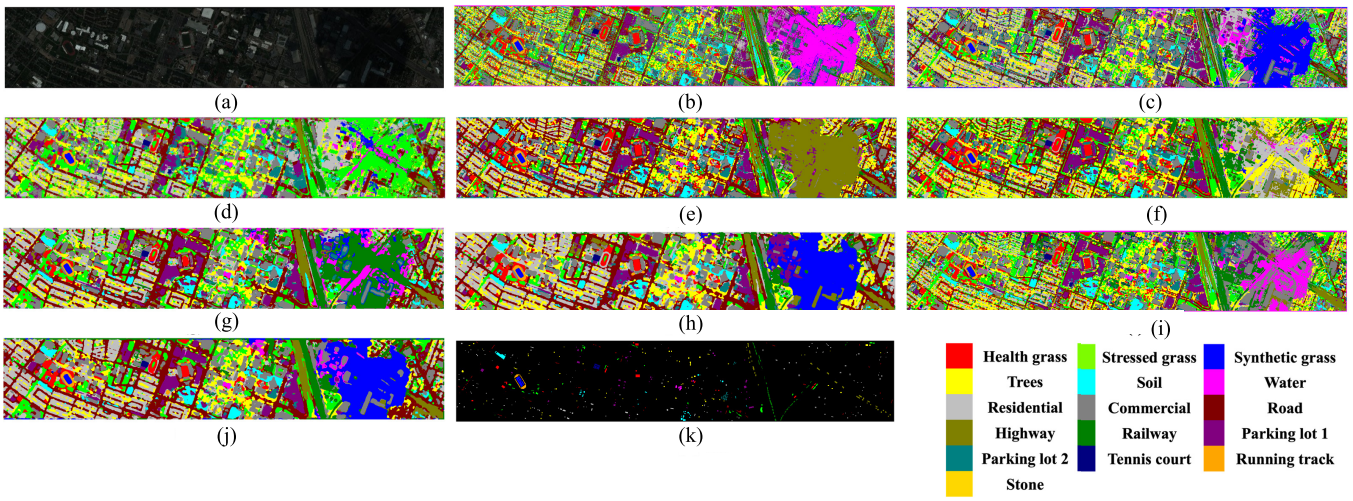


Fig. 17. Visualization of classification maps generated by different methods on the HU dataset. (a) False color image. (b) SVM-Ref. (c) CDCNN. (d) SMG-EG. (e) A2S2K. (f) SSRN. (g) DBMA. (h) DBDA. (i) ViT. (j) Proposed. (k) GT.

TABLE X
ABLATION STUDY ON ATTENTIONAL MODULES

Attention		SA	SA+SI	SA+PE	SA+PE+SI
OA	IN	92.72±0.43	93.40±0.32	93.17±0.62	**94.29±0.56**
	UP	96.10±0.56	96.43±0.59	96.83±0.36	**97.03±0.69**
	BS	95.17±1.11	95.49±0.52	95.56±0.60	**96.73±0.51**
AA	IN	90.79±1.32	91.08±1.14	**91.09±1.28**	91.26±1.13
	UP	96.05±0.57	**96.48±0.45**	96.32±0.76	96.39±0.93
	BS	96.01±0.69	96.36±1.01	96.24±0.35	**96.96±0.57**
Kappa	IN	91.68±0.49	92.46±0.38	92.20±0.72	**93.49±0.64**
	UP	94.81±0.75	95.25±0.10	95.78±0.48	**96.40±0.93**
	BS	95.11±0.56	95.36±1.21	95.19±0.66	**96.46±0.55**

TABLE XI
ABLATION STUDY ON POSITIONAL ENCODING

Positional Encodings		Absolute Encoding	Relative Encoding
OA	IN	93.39±0.24	**94.29±0.56**
	UP	96.63±0.31	**97.03±0.69**
	BS	95.56±0.84	**96.73±0.51**
AA	IN	90.44±1.14	**91.26±1.13**
	UP	96.26±0.56	**96.39±0.93**
	BS	96.40±0.73	**96.96±0.57**
Kappa	IN	93.03±0.28	**93.49±0.64**
	UP	95.52±0.42	**96.39±0.93**
	BS	95.27±0.91	**96.46±0.55**

and spectral information. However, among the experimental datasets, IN and UP datasets contain more complex land-cover features. Therefore, for the scattered classes such as the oats and grass-pasture-mowed, the training pixels are extremely limited, only one pixel is selected in the training process, and the network misclassifies them to the classes of the surrounding large area. Therefore, the AA value with all attention modules is slightly worse than that with only SA attention modules. However, overall, the adopted attention modules enhance the fusion accuracy and greatly improve the classification performance. It is worth noting that a significant improvement in terms of Kappa coefficient indicates a better class balance, which indicates the proposed method.

4) Ablation Study on Positional Encoding: In the proposed BS2T, we adopt the 2-D relative positional encoding, and compared with the absolute positional encoding, the relative positional encoding can better depict the image-to-image spatial content positions. To demonstrate the effectiveness of the relative positional encoding, we fix all the other parameters to compare the classification results between using the relative positional encoding with the absolute positional encoding. Table XI shows the classification results on the three randomly selected training datasets using the relative position encoding and the absolute positional encoding. Significant improvement can be observed by using the relative positional encoding because relative distances between features at different locations are involved to associate information across objects with positional awareness.

H. Discussion

Experimental results with randomly selected training samples and disjoint training samples both demonstrate the superiority of our proposed method when compared with SOTA deep learning-based HS image classifiers. By introducing the spatial–spectral self-attention mechanism and positional encoding, the proposed BS2T can effectively depict the long-range global dependency of the HS images. By analyzing the experimental results, the global dependency can be reflected in the following aspects.

1) *In Comparison Against Other Methods:* The proposed classification framework followed the HS spatial–spectral feature extraction paradigm. Compared with the classical spatial–spectral feature extraction paradigm methods, the proposed method achieves a significant improvement both subjectively and objectively, which indirectly demonstrates the strong global information grasp ability of the transformer. Moreover, for the disjoint datasets, the improvement is more obvious, especially when the training sample is extremely limited.

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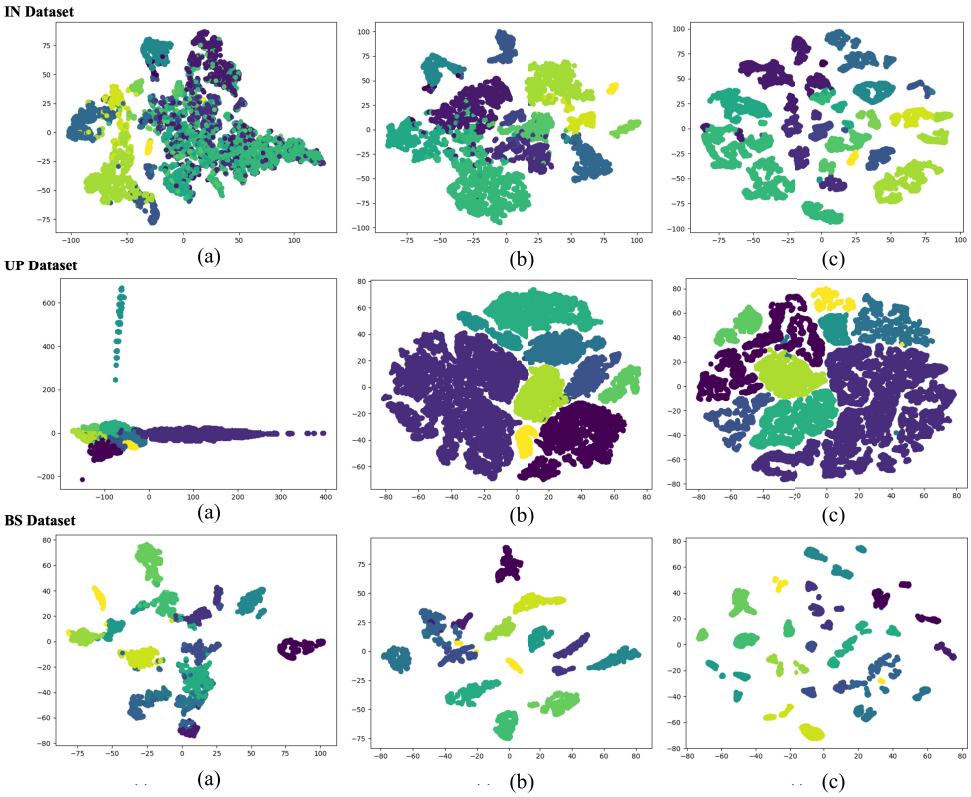


Fig. 18. t-SNE visualization of the learned features in the latent space generated by BS2TNET on the three random selected datasets. (a) CDCNN. (b) SSRN. (c) BS2TNET.

2) *In Comparison by Varying the Training Samples:* For ablation study on the training samples, as shown in Fig. 10 and Tables IV–VI, it can be observed that the proposed method shows obvious improvement in the general trend, but when the training sample is limited, the improvement is more significant. When the training sample is limited, the long-range global dependency of the HS images enhances the feature discrimination ability of each class. Meanwhile, for ablation study on attentional modules, in the proposed framework, we combine the traditional self-attention modules with positional information and spectral information. From Table IX, significant improvement can be observed by applying positional encoding and spectral information.
3) Fig. 18 correspondingly gives the t-SNE visualization of the learned features in the latent space generated by BS2TNET on the three randomly selected datasets. From Fig. 18, it is clear to observe that, compared with the CNN-based method including CDCNN and SSRN, by introducing the proposed BS2T, the network is conductive to generate more discriminative features, which indicates that the proposed BS2T can effectively depict the long-range global dependency of the HS images.

V. Conclusion

In this article, we propose a novel BS2T for HS image classification. In BS2T, to make the attention more positional aware and spectral aware, the content positional coding and spectral information is introduced into MHSA, and the novel MHS2A is proposed to depict the long-range dependency of HS pixels. On this basis, a dual-branch HS image classification framework based on 3-D CNN and BS2T is proposed for the joint local–global feature extraction. For each branch, the original HS data cubes are initially fed into dense 3-D CNN feature extraction backbones to obtain the global features, and the proposed BS2T blocks are then used to capture the global long-range band and pixel dependencies of the HS data cubes. Finally, the obtained HS spatial–spectral features are fused to get the classification result. Experimental results on four public HS image classification datasets demonstrate the superiority of our proposed method when compared with SOTA deep learning-based HS image classifiers.

The proposed BS2T provides reference significance for other HS image processing tasks, such as HS image change detection and HS image anomaly detection. In future work, we will continue to explore a more general transformer-based classification framework to deal with HS image classification tasks under complex conditions.

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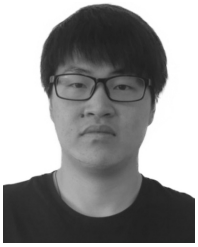
Ruoxi Song received the Ph.D. degree from the School of Geography, Liaoning Normal University, Dalian, China, in 2021.

She currently holds a post-doctoral position at the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. Her research interest is related to multitemporal analysis of hyperspectral data.



Yining Feng received the M.Eng. degree from the School of Computer and Information and Technology, Liaoning Normal University, Dalian, China, in 2022. He is currently pursuing the Ph.D. degree with the School of Geography, Liaoning Normal University.

His research interests include multimodal remote sensing image classification.



Wei Cheng received the B.Sc. degree from the School of Computer and Information and Technology, Liaoning Normal University, Dalian, China, in 2020, where he is currently pursuing the M.Sc. degree.

His research interest includes multimodal remote sensing image change detection.



Zhenhua Mu received the M.Sc. degree from the School of Computer and Information and Technology, Liaoning Normal University, Dalian, China, in 2020. She is currently pursuing the Ph.D. degree with the School of Geography, Liaoning Normal University.

Her research interest is related to anomaly detection of hyperspectral image.



Xianghai Wang received the Ph.D. degree from Jilin University, Changchun, China, in 1999.

He is currently a Professor with the School of Computer and Information Technology, Liaoning Normal University, Dalian, China. His research interests include image processing, multimedia information processing, and remote sensing image processing.

Dr. Wang is a Senior Member of the China Computer Federation (CCF).

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